Android Developer Fundamentals V2

Location-Based Services

Lesson 10



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Objectives

- Using the emulator to test location-based functionality
- Understanding forward and reverse geocoding
- Creating interactive maps with Map Views and Map Activities
- Finding your location with location-based services
- Using proximity alerts

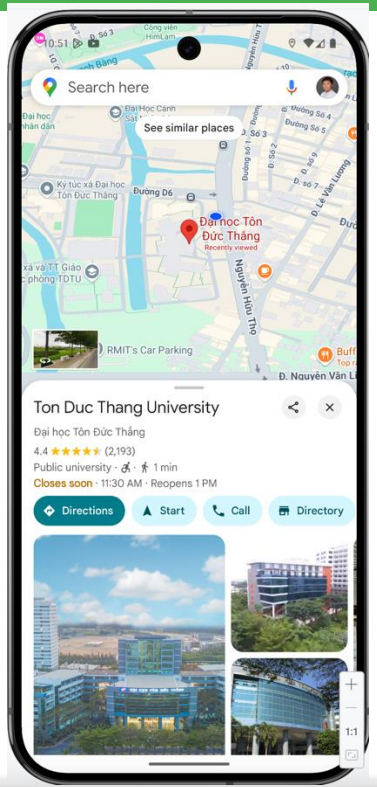
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Objectives

- Changing the map camera position
- Displaying user location on a map
- Adding markers, shapes, and image overlays to maps
- Adding awareness of the user's context using awareness snapshots
- Setting and monitoring contextual awareness fences

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Location-based functionality using emulator



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Legacy location-based services

- **Location Manager**
  - Provide hooks to the location-based services
- **Location Providers**
  - Provide location-finding technology used to determine the device's current location

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Location Manager

- Obtain your current location
- Follow movement
- Set proximity alerts for detecting movement into and out of a specified area
- Find available Location Providers
- Monitor the status of the GPS receiver

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Location-based Services permissions

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name=
    "android.permission.ACCESS_BACKGROUND_LOCATION" />
```

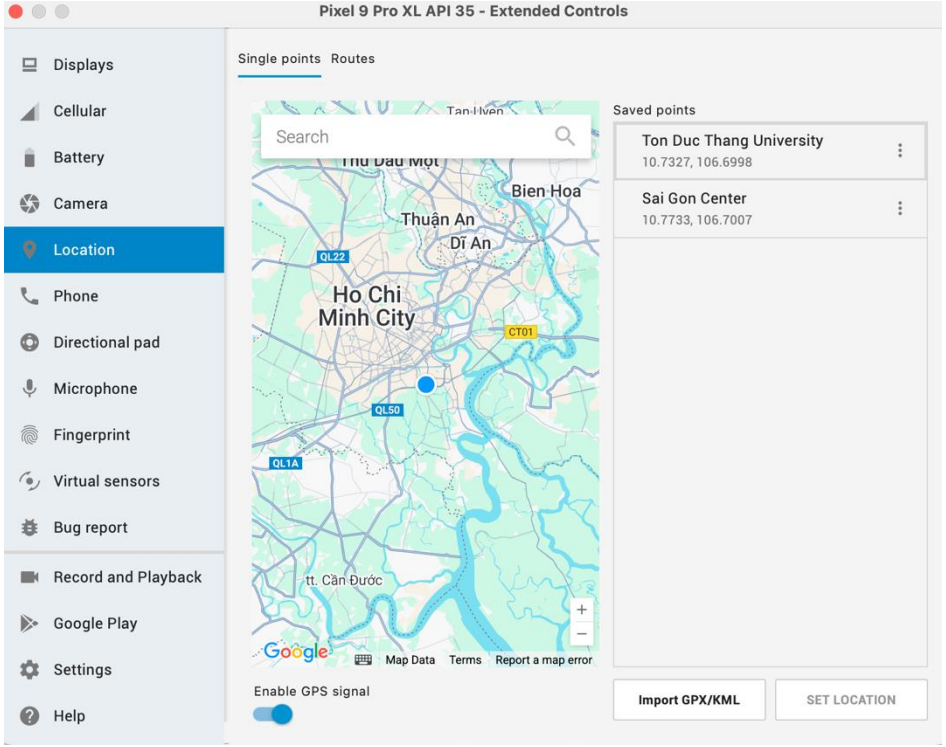

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Request LBS permissions

```
if (ActivityCompat.checkSelfPermission(this, android.Manifest.permission.ACCESS_FINE_LOCATION) !=
PackageManager.PERMISSION_GRANTED
        && ActivityCompat.checkSelfPermission(this,
android.Manifest.permission.ACCESS_COARSE_LOCATION) != PackageManager.PERMISSION_GRANTED)
{
    ActivityCompat.requestPermissions(this, new String[] {
                android.Manifest.permission.ACCESS_FINE_LOCATION,
                android.Manifest.permission.ACCESS_COARSE_LOCATION },
            CODE_PERMISSION_LOCATION);
}
```

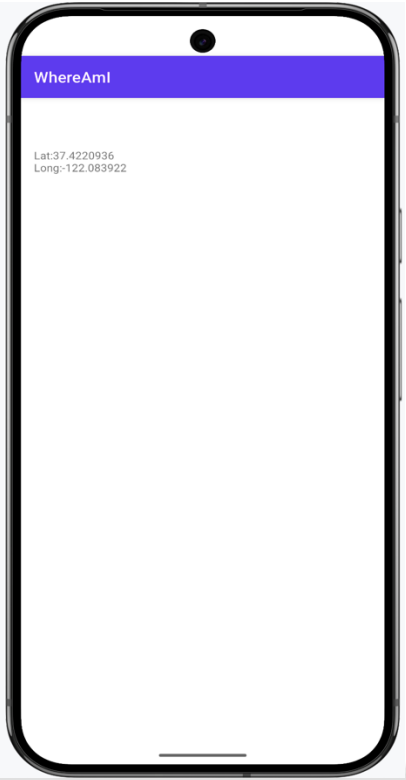

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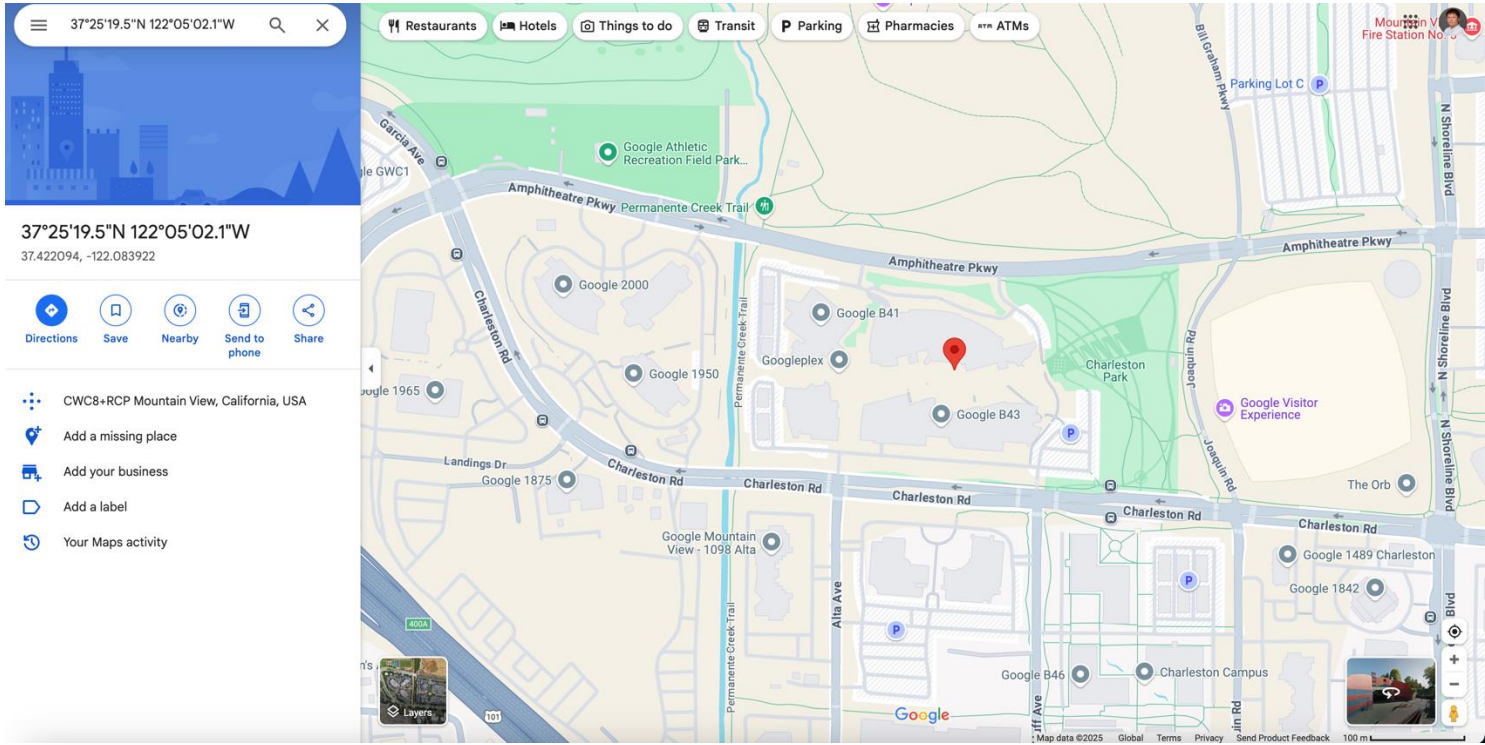
Updating Locations in Emulator Location Providers



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“Where Am I”: Find location





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Where am I: Service availability

```
protected void onCreate(Bundle savedInstanceState) {    super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        mTextView = findViewById(R.id.myLocationText);
        GoogleApiAvailability availability = GoogleApiAvailability.getInstance();
        int result = availability.isGooglePlayServicesAvailable(this);
         if (result != ConnectionResult.SUCCESS) {
                if (!availability.isUserResolvableError(result)) {
                                Toast.makeText(this, ERROR_MSG, Toast.LENGTH_LONG).show();
                }
        }
}
```

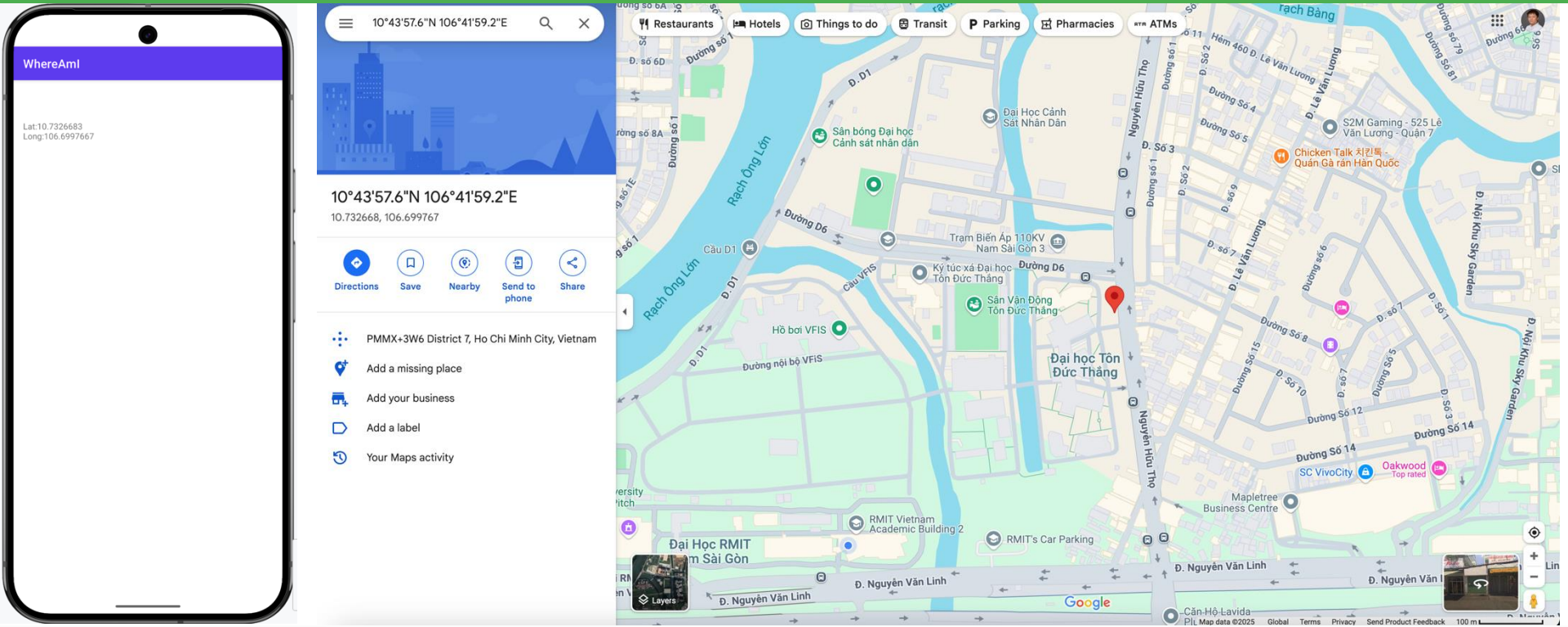

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Where am I: Get last location

```
private void getLastLocation() {
        FusedLocationProviderClient fusedLocationClient =LocationServices.getFusedLocationProviderClient(this);
        if (ActivityCompat.checkSelfPermission(this, ACCESS_FINE_LOCATION == PERMISSION_GRANTED ||
            ActivityCompat.checkSelfPermission(this, ACCESS_COARSE_LOCATION == PERMISSION_GRANTED) {
            fusedLocationClient.getLastLocation()
               .addOnSuccessListener(this, new OnSuccessListener<Location>() {
                  @Override
                  public void onSuccess(Location location) {
                     updateTextView(location);
                  }
               });
        }
}
```


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Where Am I: Update Location



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Where am I: Listen for location changes

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_where_am_iactivity);
    mTextView = findViewById(R.id.myLocationText);
    GoogleApiAvailability availability = GoogleApiAvailability.getInstance();
    int result = availability.isGooglePlayServicesAvailable(this);
    if (result != ConnectionResult.SUCCESS) {
        if (!availability.isUserResolvableError(result)) {
            Toast.makeText(this, ERROR_MSG, Toast.LENGTH_LONG).show();
        }
    }
    mLocationRequest = new LocationRequest()
            .setInterval(5000)
            .setPriority(LocationRequest.PRIORITY_HIGH_ACCURACY);
}
```


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Where am I: Update location

```
LocationCallback mLocationCallback = new LocationCallback() {
    @Override
    public void onLocationResult(LocationResult locationResult) {
        Location location = locationResult.getLastLocation();
        if (location != null) {
            updateTextView(location);
        }
    }
};
```


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Where am I: Check location settings

```
LocationSettingsRequest.Builder builder = new LocationSettingsRequest.Builder()
                .addLocationRequest(mLocationRequest);
SettingsClient client = LocationServices.getSettingsClient(this);
Task<LocationSettingsResponse> task = client.checkLocationSettings(builder.build());
task.addOnSuccessListener(this,
        new OnSuccessListener<LocationSettingsResponse>() {
          @Override
          public void onSuccess(LocationSettingsResponse
                                locationSettingsResponse) {
            // Location settings satisfy the requirements of the Location Request.
            // Request location updates.
            requestLocationUpdates();
          }
        });
```


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Where am I: Update changes

```
protected void onActivityResult(int requestCode, int resultCode,Intent data) {
        super.onActivityResult(requestCode, resultCode, data);
        final LocationSettingsStates states = LocationSettingsStates.fromIntent(data);
        if (requestCode == REQUEST_CHECK_SETTINGS) {
          switch (resultCode) {
            case Activity.RESULT_OK:
              // Requested changes made, request location updates.
              requestLocationUpdates();
              break;
            case Activity.RESULT_CANCELED:
              // Requested changes were NOT made.
              Log.d(TAG, "Requested settings changes declined by user.");
              // Check if any location services are available, and if so request location updates.
              if (states.isLocationUsable()) requestLocationUpdates();
              else Log.d(TAG, "No location services available.");
              break;
            default: break;
          }
        }
}
```


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Best practices when using location

- Battery life vs accuracy
- Minimize update rate
- Modify the fastest interval
- Unsubscribe when appropriate

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Setting and managing geofences

- Geofences are defined by a given latitude and longitude, combined with an effective radius
- Set Pending Intents that are fired based on the users' proximity to specified locations
- Can specify up to 100 Geofences per device user

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Using the geocoder

- **Forward geocoding:** Finds the latitude and longitude of an address
- **Reverse geocoding:** Finds the street address for a given latitude and longitude

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Forward geocoding vs reverse geocoding

- **Forward geocoding**
  - **Address parsing:** The input address is broken into components (e.g., house number, street name, city, and postcode).
  - **Data matching:** These components are matched against a database of geocoded addresses and corresponding coordinates.
  - **Interpolation:** If an exact match isn't found, the system may estimate the location using nearby addresses and street network data.

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Forward geocoding vs reverse geocoding

- **Use cases of forward geocoding**
  - **Mapping and navigation:** Displaying locations on maps and calculating routes
  - **Location-based services:** Enabling ride-hailing, food delivery, and store locators
  - **Business analysis:** Optimizing delivery routes and targeting location personalization
  - **Emergency response:** Locating incidents and dispatching services quickly

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Forward geocoding vs reverse geocoding

- **Reverse geocoding**
  - **Coordinate input :** Latitude and longitude values initiate the process
  - **Database search:** These coordinates are compared against a geographic database
  - **Nearest neighbor identification:** The system identifies the closest relevant feature
  - **Output generation:** A human-readable location description is provided

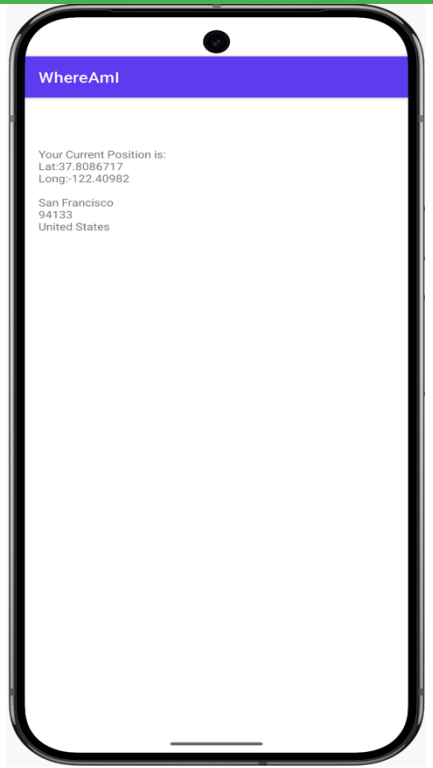
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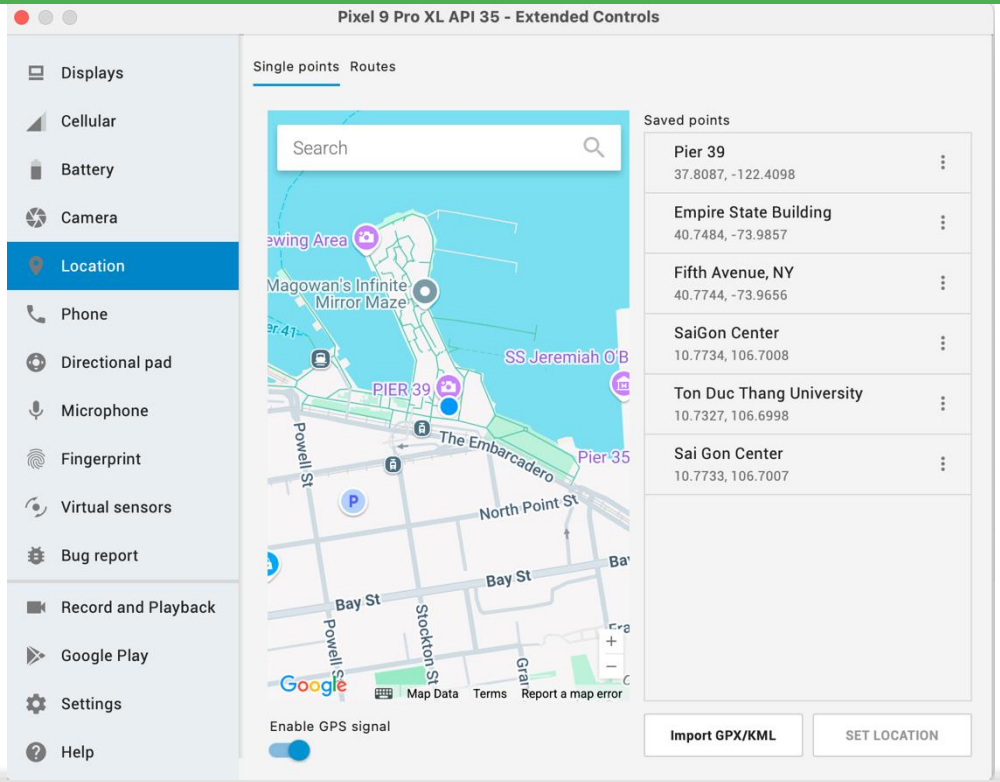
Forward geocoding vs reverse geocoding

- **Use cases of reverse geocoding**
  - **Location-aware apps:** Displaying user location or nearby POI
  - **Asset tracking:** Determining the location of vehicles or equipment
  - **Data analysis:** Understanding geographic patterns and trends
  - **Emergency services:** Locating the origin of 911 calls

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Where Am I: Geocoding





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Where am I: Geocode location

```
private String geocodeLocation(Location location) {
    String returnString = "";
    if (location == null || !Geocoder.isPresent()) return returnString;
    else {
        Geocoder gc = new Geocoder(this, Locale.getDefault());
        try {
            List<Address> addresses = gc.getFromLocation(location.getLatitude(), location.getLongitude(), 1); // One Result
            StringBuilder sb = new StringBuilder();
            if (addresses.size() > 0) {
                Address address = addresses.get(0);
                for (int i = 0; i < address.getMaxAddressLineIndex(); i++) sb.append(address.getAddressLine(i)).append("\n");
                sb.append(address.getLocality()).append("\n"). append(address.getPostalCode()).append("\n").append(address.getCountryName());
            }
            returnString = sb.toString();
        } catch (IOException e) {
            Log.e(TAG, "I/O Error Geocoding.", e);
        }
        return returnString;
    }
}
```


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Creating map-based activities

- Provide context for a physical location or address
- Use GoogleMap from within a MapFragment to create Activities that include an interactive map
- Google Maps support annotation using markers, shapes, and image overlays that can be pinned to geographical locations

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Google API credentials

https://console.developers.google.com/apis/credentials

```
<meta-data
    tools:replace="android:value"
    android:name="com.google.android.gms.version"
    android:value="@integer/google_play_services_version" />
<meta-data
    android:name="com.google.android.geo.API_KEY"
    android:value=”xxxxxxxxxx" />
```

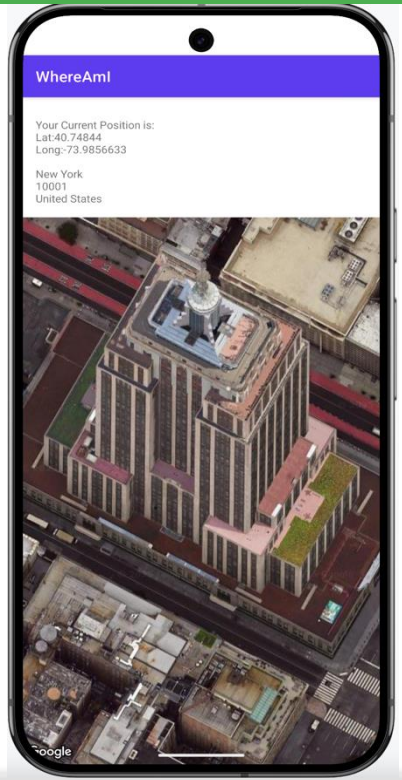

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Request a reference to the Google Map

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_where_am_iactivity);
    mTextView = findViewById(R.id.myLocationText);
    // Obtain the SupportMapFragment and request the Google Map object.
    SupportMapFragment mapFragment = (SupportMapFragment)getSupportFragmentManager().findFragmentById(R.id.map);
    mapFragment.getMapAsync(this);
    GoogleApiAvailability availability = GoogleApiAvailability.getInstance();
    int result = availability.isGooglePlayServicesAvailable(this);
    if (result != ConnectionResult.SUCCESS) {
        if (!availability.isUserResolvableError(result)) {
            Toast.makeText(this, ERROR_MSG, Toast.LENGTH_LONG).show();
        }
    }
    mLocationRequest = new LocationRequest().setInterval(5000) .setPriority(LocationRequest.PRIORITY_HIGH_ACCURACY);
}
```


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Where Am I: Map-based activities



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References

Meier, R. (2012). *Professional Android 4 application development*. John Wiley & Sons.

Meier, R., & Lake, I. (2018). *Professional Android*. John Wiley & Sons.

Team, R. (2024). *Forward geocoding vs. reverse geocoding*. https://radar.com/blog/forward-geocoding-vs-reverse-geocoding#:~:text=Comparing%20forward%20and%20reverse%20geocoding,a%20location%20description%20from%20coordinates.

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